

Stafny Velitia Karkada

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TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++

Frontend: React, Redux(Saga/Thunk), React Native, Vue, Next.js, HTML5, CSS, SASS, ES6, CMS(Contentful,Drupal), Bootstrap, Jest, Tailwind, Enzyme, React Testing Library, Cypress, Playwright, axe-core, WAVE, MUI, Figma, Deque, WCAG 2.2

Backend & Infra: Node.js, Express, REST APIs, GraphQL, PostgreSQL, MongoDB, Firebase, Oracle DB, SQL, Neo4j, FastAPI, Docker, Kubernetes, AWS, GCP, Jenkins, GitHub Actions, OAuth 2.0, JWT, Google Maps Platform, Jira, Confluence

AI & ML: RAG, PineCone, TensorFlow, PyTorch, RNNs, Transformers, LangChain, SSE, Amazon Bedrock, Lovable, Bolt

EXPERIENCE

Senior Software Engineer, Dune Security (NYC, USA)

Oct 2025 - Present

- Rebuilt Dune Security's frontend from JavaScript to TypeScript Monorepo using Next.js. Introduced type-safe workflows, pre-commit validation hooks, TanStack Query-based client-side caching, ephemeral Vercel environments, and unit testing with Vite, reducing recurring production bugs from 20+ monthly to near-zero.
- Led backend migration from a monolith to microservices, delivering 10+ Go-based APIs and modularizing application architecture; built event-driven workflows with AWS SQS and Lambda, automated report generation and email delivery through n8n and SES, and optimized MongoDB schema design and indexing.
- Engineered AI-agentic phishing using Claude-based persona agents for adaptive SMS, Viber, and RCS campaigns, integrating Redis queues, DynamoDB session storage, and OpenWebUI for prompt testing while supporting thousands of concurrent users.
- Built an LLM-assisted PR review workflow integrating Claude API and CodeRabbit into GitLab pipelines, to standardize review outputs, reduce manual review overhead, and accelerate developer feedback cycles across teams.

UX Engineer, Autodesk (SF, USA)

Apr 2024 - Dec 2024

- Developed a team-management workflow with React, Drupal, and Material UI, enabling role-based access control and centralized state management to improve maintainability, and increase developer engagement by 60% (Pendo).
- As Autodesk's first Accessibility-focused Engineer, identified recurring accessibility issues across all products, traced failures to shared design-system, and partnered with engineering, design, and platform teams to drive upstream fixes, author centralized standards, and led technical workshops that scaled WCAG 2.2 adoption across the organization.
- Led Accessibility audit of the Autodesk Developer Portal, identified 100+ issues using Axe DevTools, WAVE, and Accessibility Insights, and fixed violations with axe-core, axe-jest, Playwright, and Cypress.

Software Engineer, Y Media Labs (Bangalore, India)

Mar 2021 - Sep 2023

- Revamped California Pizza Kitchen's web platform using React Hooks(TypeScript), lazy loading, cutting user clicks by 75%, boosting conversions 2.5x, doubling loyalty memberships. Enhanced SEO and Core Web Vitals using Lighthouse audits and CI-integrated metrics reporting, recognized by their CMO.
- Built a full-stack white-labeled B2B web solution for Clover Payments using React, Node.js, and MongoDB, enabling rebranding for 20+ clients. Wrote unit tests with Jest and React Testing Library to ensure reliability.
- Partnered directly with client teams and internal stakeholders across multiple projects to run discovery sessions, translate business goals into product direction, and rapidly iterate on Figma prototypes to align design and engineering before execution.
- Built YML's first design system, developed reusable React components and design tokens, standardizing UI patterns, increasing production efficiency by 70% and reducing team size requirements by 60%.
- Led early GPT-3 (2021) integration on the company website, architecting Python FastAPI backend services for real-time AI features via API integration, enhancing user interaction and client-facing discovery of company work.

Associate Software Engineer, Y Media Labs (Bangalore, India)

May 2020 - Mar 2021

- Deployed a data-heavy website using Redux Sagas for state management, implemented efficient pagination and client-side caching to optimize load times. Set up SonarQube in the Jenkins CI/CD pipeline for code quality checks.
- Built a CMS-integrated Vue.js site for WeTransfer using GraphQL and Contentful, enabling the marketing team to manage content independently using headless CMS, reducing developer dependency and accelerating time-to-market.
- Built an in-house React Native mobile app for health tracking, using Firebase Firestore, Firebase Cloud Messaging (FCM) for push notifications, and Google Analytics to track feature adoption and engagement metrics.

Graduate Web Teaching Assistant, Northeastern University (Boston, USA)

Jan 2024 - Aug 2025

- Delivered live coding workshops and provided real-time debugging support while mentoring 140+ students and 50+ project teams in building end-to-end, multi-user applications with WebSockets, JWT authentication, PostgreSQL and AI integrations.
- Guided students in designing RESTful platforms with PostgreSQL schema management using re-executable DDL/DML scripts, PL/pgSQL procedures, database triggers, and AWS-based CI/CD deployments following Twelve-Factor App principles.

PROJECTS & ACADEMIC RESEARCH

Wyckoff AI Trading Bot | Next.js, TypeScript, Tailwind CSS, Python, LLMs

- Built a Python-based AI trading assistant using FastAPI, OpenAI models, and RAG pipelines with vector embeddings to deliver conversational trading workflows and implemented real-time streaming architecture with Server-Sent Events (SSE).
- Experimented with Neo4j Bloom and Amazon Bedrock to enhance chatbot capabilities, leveraging Neo4j Knowledge Graph to model trading concepts and deliver graph-based insights.

EDUCATION

MS in Computer Science: Northeastern University (Boston, MA, USA)

May 2025

Bachelor of Engineering in Computer Science: NMAM Institute Of Technology (Karnataka, India)

Aug 2020